

E567736_25

The equation $x^4 + xy + y^3 - 2y + 3 = 0$ has no integer solutions

Size $h = 35$, length $l = 12.58$

Theorem 1. *The equation*

$$x^4 + xy + y^3 - 2y + 3 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), with $(d_1, d_2) = (15, 15)$, the fact that an odd prime p with $(-d_k/p) = -1$ divides $U_k^2 + d_k V_k^2$ only to an even power, and the product formula for Hilbert symbols. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols, Hilbert symbols and the norm forms $a^2 + db^2$

Throughout this section, d is an odd squarefree integer with $d \neq -1$; we apply the results with $d \in \{15\}$. For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , it depends only on a modulo p , and $(-1/p) = (-1)^{(p-1)/2}$ [1, Chapter 5].

Lemma 2. *Let p be an odd prime with $(-d/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + db^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv -d(bc)^2 \equiv -d \pmod{p}$, so $-d$ is a square modulo p . As $(-d/p) \neq 0$, it is a non-zero square, so $(-d/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 + db^2 \neq 0$, and $v_p(a^2 + db^2)$ is even for every odd prime p with $(-d/p) = -1$.*

Proof. If $d > 0$, then clearly $a^2 + db^2 > 0$. If $d < 0$, then $-d > 1$ is squarefree, hence not a square, so $a^2 + db^2 \neq 0$. Let p be an odd prime with $(-d/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + db'^2$ by Lemma 2. Since $a^2 + db^2 = p^{2k}(a'^2 + db'^2)$, we get $v_p(a^2 + db^2) = 2k$. $\square$

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. For an odd integer t , we put $\varepsilon(t) = (t-1)/2$ and $\omega(t) = (t^2-1)/8$. We use the following facts [2, Chapter III, Theorems 1 and 3].

Theorem 4. *Let a and b be non-zero integers.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.

- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\epsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\epsilon(u)\epsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.

Lemma 5. Let n be a non-zero integer.

- (a) If $n > 0$ or $d < 0$, then $(n, -d)_\infty = 1$.
- (b) If p is a prime with $p \nmid 2d$, then $(n, -d)_p = (-d/p)^{v_p(n)}$.
- (c) If p is a prime with $p \mid d$, write $d = pd'$ and $n = p^j u$ with $p \nmid u$. Then $(n, -d)_p = (u/p)(d'/p)^j$.
- (d) Write $n = 2^j u$ with u odd. Then $(n, -d)_2 = (-1)^{\epsilon(u)\epsilon(-d) + j\omega(d)}$. In particular, $(n, -d)_2 = 1$ if $-d \equiv 1 \pmod{8}$.

Proof. Part (a) follows from Theorem 4(i), as $-d < 0$ only if $d > 0$. For (b) and (c), we apply Theorem 4(ii) with $a = n$ and $b = -d$. If $p \nmid d$, then $\beta = 0$ and $w = -d$, which gives (b). If $p \mid d$, then $\beta = 1$ and $w = -d'$, which gives $(n, -d)_p = (-1)^{j\epsilon(p)}(u/p)(-d'/p)^j = (u/p)(d'/p)^j$, because $(-1/p) = (-1)^{\epsilon(p)}$. Part (d) is Theorem 4(iii) with $a = n$, $b = -d$, $\beta = 0$ and $w = -d$, because $\omega(-d) = \omega(d)$. If $-d \equiv 1 \pmod{8}$, then $\epsilon(-d)$ is divisible by 4 and $d^2 \equiv 1 \pmod{16}$, so both exponents are even. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + xy + y^3 - 2y + 3$, so that (1) is the equation $F(x, y) = 0$. For the first identity, $d_1 = 15$, and we define

$$\begin{aligned}
H_1 &= x^2 - 9x - y + 14, \\
T_1 &= 4x^4 - 72x^3 + 4x^2y + 436x^2 - 36xy - 1004x + 4y^2 + 56y + 776, \\
U_1 &= 2x^3 - 27x^2 + 100x - 104, \\
V_1 &= x^2 - 4x + 2, \\
G_1 &= -4, \\
A_1 &= 10x - 7, \\
B_1 &= -20x^2 + 204x - 333,
\end{aligned}$$

for the second identity, $d_2 = 15$, and we define

$$\begin{aligned}
H_2 &= x^2 - y - 1, \\
T_2 &= 4x^4 + 4x^2y - 8x^2 + 4x + 4y^2 - 4y - 4, \\
U_2 &= 2x^3 - 2x + 1, \\
V_2 &= 1, \\
G_2 &= -4, \\
A_2 &= 0, \\
B_2 &= 1,
\end{aligned}$$

and, for the signs, we also use

$$Q_1 = x^4 - 18x^3 + x^2y + 109x^2 - 9xy - 251x + y^2 + 14y + 194,$$

$$\begin{aligned}
M_1 &= x^6 - 27x^5 + 286x^4 - 1484x^3 + 3979x^2 - 5260x + 2719, \\
E_1 &= 3x^4 - 54x^3 + 327x^2 - 752x + 580, \\
Q_2 &= x^4 + x^2y - 2x^2 + x + y^2 - y - 1, \\
M_2 &= x^6 - 2x^4 + x^3 + x^2 - x + 4, \\
E_2 &= 3x^4 - 6x^2 + 4x - 5.
\end{aligned}$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 - 15V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 - 15V_2^2 = G_2F, \quad (3)$$

$$A_1U_1 + B_1V_1 = 2 \cdot 31, \quad (4)$$

$$A_2U_2 + B_2V_2 = 1, \quad (5)$$

$$-F = Q_1H_1 - M_1, \quad (6)$$

$$4Q_1 = (x^2 - 9x + 2y + 14)^2 + E_1. \quad (7)$$

$$-F = Q_2H_2 - M_2, \quad (8)$$

$$4Q_2 = (x^2 + 2y - 1)^2 + E_2. \quad (9)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) and (3) give

$$H_kT_k = U_k^2 + d_kV_k^2 \quad (k = 1, 2), \quad (10)$$

where the polynomials now denote their values at (x, y) .

Step 1: local symbols. For a prime p , let B_p be the set of k such that $p \mid d_k$, or, if $p = 2$, such that $-d_k \not\equiv 1 \pmod{8}$, and let S be the set of primes p with $B_p \neq \emptyset$. Here $S = \{3, 5\}$, with $B_3 = \{1, 2\}$ and $B_5 = \{1, 2\}$. For $p \in S$ and $k \in B_p$, we write $H_k = p^{j_k}u_k$ with $p \nmid u_k$, and $p^j \cdot u$ for the class of the integers p^ju' with $p \nmid u'$ and $u' \equiv u \pmod{p}$, or $u' \equiv u \pmod{8}$ if $p = 2$. By Lemma 5(c) and (d), the class of H_k determines $(H_k, -d_k)_p$.

The prime 3. Checking the 3^2 residue pairs modulo 3, we find that (1) has exactly 2 solutions modulo 3, and that at each of them $3 \nmid H_k$ for $k \in B_3$, so that the residue of (H_1, H_2) modulo 3 determines its class. The classes of (H_1, H_2) that occur, and the values of $((H_k, -d_k)_3)_{k \in B_3}$, are $(1, 1) \mapsto (1, 1)$ and $(2, 2) \mapsto (-1, -1)$. Hence $((H_k, -d_k)_3)_{k \in B_3}$ is $(1, 1)$ or $(-1, -1)$.

The prime 5. Checking the 5^2 residue pairs modulo 5, we find that (1) has exactly 2 solutions modulo 5, and that at each of them $5 \nmid H_k$ for $k \in B_5$, so that the residue of (H_1, H_2) modulo 5 determines its class. The classes of (H_1, H_2) that occur, and the values of $((H_k, -d_k)_5)_{k \in B_5}$, are $(4, 2) \mapsto (1, -1)$ and $(4, 3) \mapsto (1, -1)$. Hence $((H_k, -d_k)_5)_{k \in B_5}$ is $(1, -1)$.

Step 2: signs. *The sign of H_1 .* Since $F(x, y) = 0$, (6) gives $Q_1H_1 = M_1$, and M_1 and E_1 depend only on x . We show that $M_1 > 0$ and $E_1 > 0$. If $x \geq 7$, write $x = 7 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_1 and E_1 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_1 > 0$ and $E_1 > 0$. It remains to consider $0 \leq x \leq 6$. For $x \in \{0, 1, 3, 4, 5, 6\}$ we have $M_1 > 0$ and $E_1 > 0$. For $x = 2$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . Hence $M_1 > 0$ and $E_1 > 0$, so (7) gives $4Q_1 \geq E_1 > 0$, and $Q_1H_1 = M_1 > 0$ gives $H_1 > 0$.

The sign of H_2 . Since $F(x, y) = 0$, (8) gives $Q_2H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -2$,

write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $-1 \leq x \leq 1$. For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = -1$ and $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . Hence $M_2 > 0$ and $E_2 > 0$, so (9) gives $4Q_2 \geq E_2 > 0$, and $Q_2H_2 = M_2 > 0$ gives $H_2 > 0$.

Step 3: primes p with $(-d_k/p) = -1$. Let $k \in \{1, 2\}$, and let p be an odd prime with $(-d_k/p) = -1$, where $(d_1, d_2) = (15, 15)$. We show that $v_p(H_k)$ is even. By (4)–(5), U_k and V_k are not both zero, so (10) and Lemma 3 show that $v_p(H_k T_k) = v_p(U_k^2 + d_k V_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 + d_k V_k^2$ by (10), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_k U_k + B_k V_k$. For $k = 1$, we have $R_1 = 2 \cdot 31$. But p is odd, and $p \neq 31$ because $(-15/31) = 1$, a contradiction. For $k = 2$, we have $R_2 = 1$. This is impossible. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_k T_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. Let $k \in \{1, 2\}$. By Step 2 and Lemma 5(a), $(H_k, -d_k)_\infty = 1$. Let p be a prime with $p \nmid 2d_k$. By Lemma 5(b), $(H_k, -d_k)_p = (-d_k/p)^{v_p(H_k)}$, which is 1 if $(-d_k/p) = 1$, and also if $(-d_k/p) = -1$, because then $v_p(H_k)$ is even by Step 3. If $-d_k \equiv 1 \pmod{8}$, then $(H_k, -d_k)_2 = 1$ by Lemma 5(d). Hence $(H_k, -d_k)_v = 1$ unless v is a prime $p \in S$ with $k \in B_p$, and Theorem 4(iv) gives $\prod_{p \in S} (H_k, -d_k)_p = 1$.

For $p \in S$, let $\sigma_p = ((H_1, -d_1)_p, \dots, (H_2, -d_2)_p)$. Its entries with $k \notin B_p$ are 1, as we have just seen, so Step 1 gives $\sigma_p \in V_p$, where

$$\begin{aligned} V_3 &= \{(1, 1), (-1, -1)\}, \\ V_5 &= \{(1, -1)\}. \end{aligned}$$

As shown above, the componentwise product $\prod_{p \in S} \sigma_p$ is $(1, 1)$. The only vector in V_5 is $(1, -1)$, so σ_3 would have to be $(1, -1)$, which is not in V_3 . This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.